# Nick Kunz

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## Research Interests

My research interests are at the intersection of data science, machine learning, and complex systems. Many of the questions I ask are motivated by understanding emergent dynamics under evolving real-world constraints. My work has direct applications in urban planning, economics, defense, and other areas of societal impact.

## Education

**Cornell University** — Ithaca, NY 2027
Ph.D., Systems Engineering, Artificial Intelligence
*Advisors: H. Oliver Gao, Francesca Parise, Wenfei Xu*

**Cornell University** — Ithaca, NY 2025
M.S., Systems Engineering
*Advisor: H. Oliver Gao*

**Columbia University** — New York, NY 2019
M.S., Urban Planning, Urban Analytics
*Advisors: Lance Freeman, Kazuki Sakamoto*

**Harvard University** — Cambridge, MA 2012
Non-Degree, Urban Planning
*Advisor: Nick Hornig*

**Arizona State University** — Tempe, AZ 2012
B.S. (Hons.), *summa cum laude*, Housing and Urban Development
*Advisors: Kevin Kellogg, Michael Underhill, Trevor Barger*

## Publications

### Journal Articles

**N. Kunz**, H. O. Gao, "A General Capacity Frontier of Complex Networks." *Preprint arXiv:2610.09574 (Under Review).* 2026. [Link]

**N. Kunz**, H. O. Gao, "Global Geolocated Realtime Data of Urban Transit Bus Idling." *Preprint arXiv:2403.03489 (Under Review).* 2026. [Link]

D. Lu, L. Cui, **N. Kunz**, *et al.*, "Urban Infrastructure and Spatiotemporal Environmental Features for EGFR-Mutant Lung Cancer." *Preprint medRxiv:18.26353481v1 (Under Review).* 2026. [Link]

C. Avci, C. Huang, **N. Kunz**, H. O. Gao, "uTECH-Population: Synthetic Population Generation with Housing Assignment." *(Under Review).* 2026.

### Conference Proceedings

W. Wu, **N. Kunz**, P. Branco, "ImbalancedLearningRegression" in *Machine Learning and Knowledge Discovery in Databases: European Conference, ECML PKDD, September 19–23, 2022, Grenoble, France,* Proceedings, Part VI, Lecture Notes in Computer Science, vol. 13718. Springer, pp. 645–648, 2023. [Link]

### Working Papers

**N. Kunz**, H. O. Gao, "A General Rate Estimator of Complex Networks." *In Preparation.* 2026.

**N. Kunz**, H. O. Gao, "Global Geolocated Historical Data of Urban Transit Bus Idling From 2024 to 2026." *In Preparation.* 2026.

**N. Kunz**, H. O. Gao, "Network Constrained Spatial Prediction: Transfer Learning Geolocations." *In Preparation.* 2026.

### Theses

**N. Kunz**, "Structural Systems Intelligence: Network-Constrained Estimation of Global Transit Dynamics," Ph.D. Dissertation, David A. Duffield College of Engineering, Cornell Univ., Ithaca, NY, 2027.

**N. Kunz**, "Unsupervised Learning for Submarket Modeling: A Proxy for Neighborhood Change," M.S. Thesis, Grad. Sch. of Arch., Plan., and Pres. (GSAPP), Columbia Univ., New York, NY, 2019. [Online]. Available: Columbia Academic Commons. [Link]

**N. Kunz**, "Realizing Interactive Architecture: A Driver of the Knowledge Economy," B.S. (Hons.) Thesis, Barrett, The Hons. Col., Arizona State Univ., Tempe, AZ, 2012. Available: Barrett, The Honors College Thesis/Creative Project Collection. [Link]

### Industry Reports

**N. Kunz**, "$n$-Dimensional Mapping by Normalized Value Approximation: Squad Immersive Virtual Trainer (SiVT) Safety Performance Metrics," Integrated Visual Augmentation System (IVAS), Microsoft, Redmond, WA, 2021. Protected from public disclosure pursuant to: Trade Secrets Act 18 USC § 1905, 5 USC § 552(b)(4), and C3.2.1.4. of US Dept. of Defense Regulation 5400.7-R. DISTRIBUTION F.

**N. Kunz**, "Normalized $n$-Dimensional Inverse Euclidean Distance: Combining Independent Performance Metrics," Integrated Visual Augmentation System (IVAS), Microsoft, Redmond, WA, 2021. Protected from public disclosure pursuant to: Trade Secrets Act 18 USC § 1905, 5 USC § 552(b)(4), and C3.2.1.4. of US Dept. of Defense Regulation 5400.7-R. DISTRIBUTION F.

**N. Kunz**, "Minimum-Weighted Mean Variability: Within Group Performance Metrics," Integrated Visual Augmentation System (IVAS), Microsoft, Redmond, WA, 2020. Protected from public disclosure pursuant to: Trade Secrets Act 18 USC § 1905, 5 USC § 552(b)(4), and C3.2.1.4. of US Dept. of Defense Regulation 5400.7-R. DISTRIBUTION F.

**N. Kunz**, "An Exploratory Data Analysis of the Gavin Cup: Bringing Readiness & Performance Metrics Into Focus," Integrated Visual Augmentation System (IVAS), Microsoft, Redmond, WA, 2020. Protected from public disclosure pursuant to: Trade Secrets Act 18 USC § 1905, 5 USC § 552(b)(4), and C3.2.1.4. of US Dept. of Defense Regulation 5400.7-R. DISTRIBUTION F.

## Software

**GRD-TRT-BUF-4I: Ground Truth Buffer for Idling** 2023 - 2027
GitHub [ Link ] arXiv [ Link ] Website [ Link ]

A novel algorithm designed as a series of scalable microservices that detects urban transit bus idling in realtime on global scale. Utilizes a combination of templated ETL pipelines to provide a comprehensive view of the emergent behavior of interfleet idling patterns across multiple transit networks around the world.

**SMOGN: Synthetic Minority Over-Sampling Technique for Regression with Gaussian Noise** 2021
GitHub [ Link ] PyPI [ Link ] Kaggle [ Link ] Stack Exchange [ Link ]

A pre-processing algorithm designed to address imbalanced data for regression problems. Conducts over-sampling with traditional interpolation, as well as with the introduction of Gaussian noise. Selects between the two over-sampling techniques by the KNN distances underlying a given observation.

**NestedHyperBoost: Nested Cross-Validation for Bayesian Optimized Gradient Boosting** 2020
GitHub [ Link ] PyPI [ Link ]

Unifies Nested K-Fold Cross-Validation, Bayesian Hyperparameter Optimization, and Gradient Boosting. Designed for rapid prototyping on small to mid-sized data sets. Quickly obtains high quality prediction results by abstracting away tedious hyperparameter tuning and implementation details in favor of usability and implementation speed.

**NestedHyperLine: Nested Cross-Validation for Bayesian Optimized Linear Regularization** 2020
GitHub [ Link ] PyPI [ Link ]

Unifies Nested K-Fold Cross-Validation, Bayesian Hyperparameter Optimization, and Linear Regularization. Designed for rapid prototyping on small to mid-sized data sets. Quickly obtains high quality prediction results by abstracting away tedious hyperparameter tuning and implementation details in favor of usability and implementation speed.

**CRASSMAT: Conditional Random Sampling Sparse Matrices** 2019
GitHub [ Link ] CRAN [ Link ]

A novel matrix sampling algorithm designed to conditionally random sample sparse matrices. Useful for training and test set splitting prior to model fitting in cross-validation procedures and for estimating the predictive accuracy of imputation methods, such as NNMF and SVD.

## Professional Experience

**Deloitte** — Seattle, WA 2021 - 2025
Senior Data Scientist
*Artificial Intelligence* [Link]

**Microsoft** — Redmond, WA 2020 - 2021
Data Scientist
*Integrated Visual Augmentation System* [Link]

**Pacific Prospecting Group** — Seattle, WA 2017 - 2019
Data Scientist
*Cryptocurrency Mining & Management* [Link]

**Brawner & Company** — Snoqualmie, WA 2016 - 2017
Development Analyst
*Real Estate Development* [Link]

**bcWORKSHOP** — Dallas, TX 2013 - 2014
Fellow
*Architecture & Planning* [Link]

## Selected Projects

**N. Kunz**, S. Huang, S. Osakata, "Cloudy Night: Activating Park Space by Illuminated Interactivity," Grad. Sch. of Arch., Plan., and Pres. (GSAPP), Columbia Univ., New York, NY, 2019. [Online]. Available:

GSAPP [ Link ] GitHub [ Link ] YouTube [ Link ] Faculty [ Link ]

**N. Kunz** and S. Bhele, "Signal or Noise? Qualified Opportunity Zone Effect on Development Activity in Queens," Grad. Sch. of Arch., Plan., and Pres. (GSAPP), Columbia Univ., New York, NY, 2019. [Online]. Available:

GSAPP [ Link ] GitHub [ Link ] Issuu [ Link ]

E. Hudgins and **N. Kunz**, "Planning for Extreme Settlements, McMurdo Station, Antarctica," Grad. Sch. of Arch., Plan., and Pres. (GSAPP), Columbia Univ., New York, NY, 2018. [Online]. Available:

GSAPP [ Link ] Issuu [ Link ]

J. Karasik, S. Lee, **N. Kunz**, *et al.*, "Negotiating Fragments (Restructuring Quarto Workshop)," Genoa, Italy, Grad. Sch. of Arch., Plan., and Pres. (GSAPP), Columbia Univ., New York, NY, 2018. [Online]. Available:

GSAPP [ Link ] YouTube [ Link ] Issuu [ Link ] Issuu [ Link ] Press [ Link ]

**N. Kunz**, "Proportionalities: A Decade of Housing Markets to Subway Stations in Manhattan from 2006 to 2016," Grad. Sch. of Arch., Plan., and Pres. (GSAPP), Columbia Univ., New York, NY, 2017. [Online]. Available:

GSAPP [ Link ] GSAPP [ Link ] Issuu [ Link ]

**N. Kunz**, "Reimagined: John Snow's Cholera Map of 1854," Grad. Sch. of Arch., Plan., and Pres. (GSAPP), Columbia Univ., New York, NY, 2017. [Online]. Available:

Issuu [ Link ]

G. S. Coch, S. Nelsen, **N. Kunz**, K. Warner, "Seeking Refuge: The Geography of Migrant Deaths Along the US-MX Border," Grad. Sch. of Arch., Plan., and Pres. (GSAPP), Columbia Univ., New York, NY, 2017. [Online]. Available:

GSAPP [ Link ] Medium [ Link ]

**N. Kunz**, "Health & Wellness Alliance for Children: Children's Asthma, Healthy Physical Environments," Dallas County, Texas, bcWORKSHOP, Dallas, TX, 2014. [Online]. Available:

Issuu [ Link ]

**N. Kunz**, "The Bexar Street Corridor: A Century in Development," Dallas, Texas, bcWORKSHOP, Dallas, TX, 2013. [Online]. Available:

Issuu [ Link ] Issuu [ Link ] bc [ Link ] Vimeo [ Link ]

**N. Kunz**, "Fenway-Kenmore Square: Comprehensive Planning & Finance," Cambridge, Massachusetts, Grad. Sch. of Design (GSD), Harvard Univ., Cambridge, MA, 2012. [Online]. Available:

Issuu [ Link ]

**N. Kunz**, "Design Representation: Analog Modeling & Hand Drafting," Grad. Sch. of Design (GSD), Harvard Univ., Cambridge, MA, 2012. [Online]. Available:

Issuu [ Link ]

**N. Kunz**, "Danelle Plaza: Complete Street Mobility Planning," Tempe, Arizona, Herberger Inst. for Des. and the Art., Arizona State Univ., Tempe, AZ, 2012. [Online]. Available:

Issuu [ Link ]

**N. Kunz**, "Ascension: Multifamily Mixed-Use Real Estate Development Proposal," Tempe, Arizona, Independent Project, Phoenix, AZ, 2012. [Online]. Available:

Issuu [ Link ]

**N. Kunz**, "Sui Residence: Second Story Architectural Design," Paradise Valley, Arizona, Independent Project, Phoenix, AZ, 2011. [Online]. Available:

Issuu [ Link ]

## Mentorship Activities

**Audrey Shin**, B.S., Computer Science, Columbia University — New York, NY 2025 - 2026
Machine Learning Engineer Intern, ORO Labs
*Professional development and management in early career engineering.*

**Albert Rosario-Pichardo**, M.S., Urban Planning, Columbia University — New York, NY 2023 - 2024
GIS Analyst Intern, New York City Economic Development Corporation
*Academic and professional development in early career public service.*

**Albert Ruiz Martinez**, B.S., Computer Science, Arizona State University — Tempe, AZ 2023 - 2024
Analyst, Deloitte Consulting
*Professional development and management in early career consulting.*

**Julia VanPutte**, B.S., Operations Research, Cornell University — Ithaca, NY 2022 - 2024
Business Intelligence Engineer Intern, Amazon
*Academic and professional development from undergraduate school to industry.*

**Chawner Hurd**, M.B.A., Business, Columbia University — New York, NY 2020 - 2023
Associate Intern, Visa
*Military transition from US Army, 5th Special Forces Group to graduate school and industry.*

**Michael Bollinger**, B.A., Computer Science, Columbia University — New York, NY 2019 - 2023
Software Engineer Intern, Amazon
*Military transition from US Army, 75th Ranger Regiment to undergraduate school and industry.*

**Daniel Hineline**, B.A., Political Science, Columbia University — New York, NY 2018 - 2023
Associate Intern, Guggenheim Partners
*Military transition from US Army, 82nd Airborne Division to undergraduate school and industry.*

## Funding & Support

### Fellowships & Scholarships

Graduate Engineering Fellowship, Cornell University — Ithaca, NY 2023
Systems Engineering Fellowship, Cornell University — Ithaca, NY 2022
Korean Ancestry Grant, William Orr Dingwall Foundation — Washington, DC 2010 - 2012
Barrett, The Honors College Scholarship, Arizona State University — Tempe, AZ 2010 - 2012
AIERF Undergraduate Scholarship, Appraisal Institute — Chicago, IL 2010 - 2011
Foundation Scholarship, Study.net — Fort Lauderdale, FL 2010
Herberger Institute for Design & the Arts Scholarship, Arizona State University — Tempe, AZ 2010
Center for Global Education Services Scholarship, Arizona State University — Tempe, AZ 2009

### Education Benefits & National Service Awards

Chapter 33: Post-9/11 GI Bill, US Dept. of Veterans Affairs — Washington, DC 2022 - 2026
Chapter 31: Veteran Readiness and Employment, US Dept. of Veterans Affairs — Washington, DC 2017 - 2019
Eli Segal AmeriCorps Education Award, US AmeriCorps — Washington, DC 2014

## Honors & Awards

**Applause Award**, Deloitte Consulting — Seattle, WA [Link] 2022
**Distinguished Alumni Award**, Arizona State University — Tempe, AZ [Link] 2016

## Talks & Presentations

**Practical Applications in Data Science**, Cornell Data Science Club, Cornell University — Ithaca, NY 2026
**Ground Truth Buffer for Idling**, 3-Minutes of Madness, Cornell University — Ithaca, NY [Link] 2024
**Global Geolocated Realtime Data**, Flash Talks, Cornell University — Ithaca, NY 2023
**Emerging Technology Workshop**, Countryside Angels Talks, Ag Ventures Alliance — Mason City, IA 2021
**Planning Paths in Private Tech**, Alumni Panel, Columbia GSAPP — New York, NY [Link] 2021
**Restructuring Quarto Workshop**, Municipality of Genoa — Genoa, IT [Link] 2018
**The Bexar Street Corridor**, US Congresswoman Eddie Bernice Johnson's Office — Dallas, TX 2014
**The Bexar Street Corridor**, TR Hoover Community Development Corporation — Dallas, TX 2014
**The Bexar Street Corridor**, bcSHOPFRONT Series, bcWORKSHOP — Dallas, TX [Link] 2013
**Realizing Interactive Architecture**, Honors Symposium, Arizona State University — Tempe, AZ 2012

## Media & Press

**Urban Planning Semester In Review: Alumni Updates**, Columbia GSAPP — New York, NY [Link] 2023
**Spotlights: Nick Kunz**, Cornell Systems Engineering — Ithaca, NY [Link] 2023
**Meet the N-K Alumni Nick Kunz**, Northwood Anchor, Vol. 126, No. 41, Pg. 4 — Northwood, IA [Link] 2022
**With Special Guest, Nick Kunz**, TWA Family Hour Podcast — Seattle, WA [Link] 2020
**Restructuring Quarto Workshop**, Municipality of Genoa Press — Genoa, IT [Link] 2018
**Achievement Oriented: Nick Kunz**, ASU Magazine, Vol. 20, No. 2, Pg. 7 — Tempe, AZ 2016
**Welcome Nick Kunz**, bcWORKSHOP — Dallas, TX 2013

## Academic Service

**Tuesday Talks Mentorship**, Columbia GSAPP — New York, NY [Link] 2026
**Alumni Mentorship Program**, Columbia GSAPP — New York, NY [Link] 2023 - 2024

## Military Service

### Assignments

**US Army, 2nd Ranger Battalion, 75th Ranger Regiment** — Fort Lewis, WA 2015 - 2016
*Infantry*

**US Army, Ranger Training Detachment** — Fort Benning, GA 2015
*Ranger Assessment and Selection Program*

**US Army, 1st Special Warfare Training Group** — Fort Bragg, NC 2014
*Special Forces Assessment and Selection*

**US Army, 1st Special Warfare Training Group** — Fort Bragg, NC 2014
*Special Forces Preparation and Conditioning*

**US Army, 507th Parachute Infantry Regiment** — Fort Benning, GA 2014
*Airborne School*

**US Army, 19th Infantry Regiment** — Fort Benning, GA 2014
*Infantry School*

### Decorations

**Expert Badge, Rifle Component Bar**, US Army, 75th Ranger Regiment — Fort Lewis, WA 2016
**Global War on Terrorism Service Medal**, US Army, 75th Ranger Regiment — Fort Lewis, WA 2016
**Driver Badge, Wheeled Vehicle Component Bar**, US Army, 75th Ranger Regiment — Fort Lewis, WA 2015
**Parachutist Badge, Unit Trimming**, US Army, 507th Parachute Infantry Regiment — Fort Benning, GA 2014
**Infantry Blue Cord Fourragère**, US Army, 19th Infantry Regiment — Fort Benning, GA 2014
**National Defense Service Medal**, US Army, 19th Infantry Regiment — Fort Benning, GA 2014
**Army Service Ribbon**, US Army, 19th Infantry Regiment — Fort Benning, GA 2014

## Civil Service

**AmeriCorps** — Dallas, TX 2013 - 2014
*Volunteer in Service to America (VISTA)*

**Open Architecture Collaborative** — Dallas, TX 2013 - 2014
*Architecture & Planning*

**Trinity Park Conservancy** — Dallas, TX 2013 - 2014
*Public Park Conservation*

## Professional Memberships

**Sigma Xi** — Research Triangle Park, NC 2026 - Present
**The New York Academy of Sciences** — New York, NY 2024 - Present
**AmeriCorps Alums of North Texas** — Dallas, TX 2014 - Present
**Barrett, The Honors College Alumni Association** — Tempe, AZ 2012 - Present
**US Military Veterans of Columbia University** — New York, NY 2017 - 2018
**Architecture for Humanity, Dallas/Fort Worth Chapter** — Dallas, TX 2013 - 2014
**Social Economic Environmental Design Network** — Princeton, NJ 2013 - 2014
**Urban Land Institute Arizona** — Phoenix, AZ 2011 - 2012
**United States Green Building Council, Central AZ Chapter** — Phoenix, AZ 2009 - 2012
**American Institute of Architecture Students** — Washington, DC 2009 - 2010

## Technical Skills

### Languages

**Scripting**: Python, Bash
**Statistical**: R, Stata
**Compiled**: C/C++, Fortran
**Database**: SQL, NoSQL, Cypher
**Typesetting**: LaTeX, Markdown

### Frameworks

**Data Science**: NumPy, SciPy, Pandas, statsmodels, Featuretools, Tidyverse
**Visualization**: Matplotlib, Seaborn, Plotly, Processing, Shiny, ggplot
**Statistical Learning**: Scikit-Learn, XGBoost, LightGBM, CatBoost
**Reinforcement Learning**: OpenAI Baselines, Stable Baselines
**AI Engineering**: Pinecone, LangChain, LangGraph
**Deep Learning**: PyTorch, TensorFlow, ONNX

### Deployment

**DevOps**: Git, Ubuntu, Docker, CI/CD
**Databases**: PostgreSQL, SQLite, CosmosDB, Neo4j
**Web Services**: Flask, Gunicorn, Nginx
**Web Automation**: Selenium, Puppeteer
**Cloud Platforms**: Azure, AWS, Databricks
**Security**: STIGs, NIST, ZAP

### Software

**Development Environments**: VS Code, RStudio, Jupyter
**Geographic Information Systems**: ArcGIS, QGIS, OSM, Leaflet, Mapbox
**3D Modeling & Optimization**: Rhino, Grasshopper, SketchUp, AutoCAD, Unity
**Design & Illustration**: Adobe Photoshop, Illustrator, InDesign
**Financial Modeling**: Excel, Macros, CoStar
**Studio & Live Audio**: Logic, Pro Tools

### Prototyping

**Project Management**: Agile, Scrum, Critical Path
**Sensors & Hardware**: Arduino, Raspberry Pi
**Electrical**: Soldering, Wiring, Safety
**Analogs**: Drawing, Hand Drafting, Physical Modeling

## Personal Activities

### Music & Live Audio

**Atlas North American Tour** — US & Canada 2013
The Word Alive *(Fearless Records)*, Co-Tour Manager
The Word Alive *(Fearless Records)*, Guitar Tech

**Fight the Silence Tour** — US 2013
The Word Alive *(Fearless Records)*, Guitar Tech
For Today *(Razor & Tie Records)*, Guitar Tech

**All Stars Tour** — US 2012
The Word Alive *(Fearless Records)*, Guitar Tech
A Skylit Drive *(Fearless Records)*, Guitar Tech

**Vans Warped Tour** — US & Canada 2011
The Word Alive *(Fearless Records)*, Co-Tour Manager
Of Mice & Men *(Rise Records)*, Guitar Tech

**Regional Performances** — US 2011
The Rendering *(Authentik Artists)*, Guitar

*Updated: Oct. 2026*